

Corporate and Datacenter Network Architecture — Guided Notes ANSWER KEY

Unit 1.7.1 | CompTIA Network+ N10-009 Obj. 1.7 | N10-008 1.7

For instructor use / distribute after students complete the notes.

Question 1

In three-tier architecture, match each layer to its primary function:

Access layer → _____

Distribution layer → _____

Core layer → _____

*Answer: Access → end devices connect here; provides port density, PoE, VLAN assignment;
Distribution → aggregates access switches, handles inter-VLAN routing, applies ACLs and policies;
Core → backbone, moves traffic between distribution blocks at wire speed with no filtering.*

Question 2

A collapsed core design merges the _____ and _____ layers into one. It is appropriate for _____ environments. It becomes a problem when _____.

Answer: Distribution and core; smaller single-building; the network grows large enough that combined switches become a bottleneck or unworkable single point of failure.

Question 3

In spine-leaf architecture, every _____ switch connects to every _____ switch. No _____ connects directly to another _____. Any server can reach any other server in exactly _____ hops.

Answer: Leaf; spine; leaf; leaf; two.

Question 4

Spine-leaf exists because data centers have predominantly _____ traffic (server-to-server), while three-tier was designed for _____ traffic (client-to-server). Three-tier forces server-to-server traffic through _____ extra hops.

Answer: East-west; north-south; multiple (up to distribution, across, back down).

Question 5

In top-of-rack switching, a switch sits _____ and servers connect with _____ cables. In end-of-row switching, a larger switch serves _____ with longer cable runs. The dominant modern approach is _____ because _____.

Answer: At the top of each rack; short (1-3 meter); multiple racks in a row; top-of-rack; short cable runs are easier to manage and it aligns naturally with spine-leaf (each ToR = a leaf).

Question 6

SDN separates the _____ plane (forwarding decisions) from the _____ plane (actual packet forwarding). A centralized _____ pushes decisions to network devices, which become simple

_____.

Answer: Control; data; SDN controller; forwarding engines.

Question 7

NFV replaces dedicated hardware _____ with software running on _____. The key advantages are _____ and _____. Examples of virtualized network functions include _____.

Answer: Appliances (firewalls, load balancers, IDS/IPS); standard server hardware (hypervisors/containers); flexibility and reduced cost; virtual firewalls, virtual load balancers, virtual WAN optimizers (any two).

Question 8 — Real World Application

REAL WORLD: You're told the data center is experiencing high latency between application servers that frequently communicate. The current design is three-tier. What architecture change would you recommend and why?

Answer: Recommend migrating to spine-leaf architecture. The problem is east-west traffic (server-to-server) being forced through multiple hops in three-tier — up to distribution, across to another distribution switch, back down. Spine-leaf reduces this to exactly two hops (up to spine, down to destination leaf) with consistent, predictable latency. Add leaf switches for server density and spine switches for bandwidth as needed without redesigning the whole network.